Supplements We say a group G is **simple** if $G \neq \{1\}$ and $\forall N \lhd G, N = \{1\}, G$.

Example Cyclic group $\mathbb{Z}/n\mathbb{Z}$, all its subgroups are normal, and are $m\mathbb{Z}/n\mathbb{Z}, m \mid n$.
Hence $Z/n\mathbb{Z}$ is simple $\iff n$ is a prime.

In short, simple cyclic group $\approx \mathbb{Z}/p\mathbb{Z}, p$ prime $\approx G, |G| = p$.

Proposition Let H, K be subgroups of G, and $K \lhd G$. $HK := \{hk \mid h \in H, k \in K\} \subset G$ is a subgroup and $HK = KH$.

Proof First, $HK = KH : \forall h \in H, k \in K, hk = \underbrace{hkh^{-1}}_{\subset hKh^{-1} = K} h \in KH$, the other side is similar.

Next, HK is a subgroup:
- $1 = 1 \cdot 1 \in HK$,
- $h_1 k_1 \cdot h_2 k_2 \in h_1 KHk_2 = h_1 HKk_2 = HK$,
- $(hk)^{-1} = k^{-1} h^{-1} \in KH = HK$. $\square$

Generalization $\forall K \leqslant G$, define $Z_G(K) := \{g \in G \mid gkg^{-1} = k\}$ / 中心化子群 and $N_G(K) := \{g \in G \mid gKg^{-1} = K\}$ / 正规化子群.

From definition $Z_G(K) \leqslant N_G(K), K \lhd N_G(K)$.

Then, if $H, K \leqslant G$ s.t. $H \leqslant N_G(K)$, then $HK = KH$ is a subgroup.

Proof Work in $N_G(K)$ where $K \lhd N_G(K)$. $\square$

Proposition $H, K \leqslant G, H \leqslant N_G(K)$, then $H \cap K \lhd H$ and

$$\begin{array}{rcl} HK/K & \simeq & H/(H \cap K) \\ hK & \leftrightarrow & h(H \cap K) \end{array}$$

Proof Let $x \in H \cap K, h \in H$. then $hxh^{-1} \in H, H \subset N_G(K) \implies hxh^{-1} \in K \implies H \cap K \lhd H$.

Consider $H \hookrightarrow HK \twoheadrightarrow HK/K$, its composite $=: f$. $f(h) = hK$.
- f is surjective: $hkK = hK = f(h)$.

$\ker(f) = H \cap K$ since $f(h) = K \iff hK = K \iff h \in K$.

Hence f induces $\overline{f} : H/(H \cap K) \simeq HK/K, \overline{f}(h(H \cap K)) = f(h) = hK$. $\square$

§ Semidirect products

Already seen: G_1, G_2 groups $\implies G_1 \times G_2$ is also a group.

Below is a "twisted" version.

External version Given H, N groups, homomorphism $\varphi : H \to \mathrm{Aut}(N)$. On the set $N \times H$, define the binary operation $(n, h)(n', h') := \left(n\varphi_{h(h')}, hh'\right)$, where $\varphi_h \in \mathrm{Aut}(N)$ is the image of h.

To check: This makes $N \times H$ into a group $N \rtimes_\varphi H$ with $1_{N \rtimes_\varphi H} = (1_N, 1_H)$,
$(n, h)^{-1} = (\varphi_{h^{-1}}(n^{-1}), h^{-1})$.

$N \hookrightarrow N \rtimes_\varphi H \hookleftarrow H$ as subgroups, and $n \mapsto (n, 1_H), h \mapsto (1_N, h), (n, 1_H)(1_N, h) = (n, h)$.
Also $N \lhd N \rtimes_\varphi H$.

Proof Idea: seek a group G s.t. $N \hookrightarrow G \hookleftarrow H, N \lhd G$, and every $g \in G$ can be expressed uniquely as $g = nh$. If so: $nh \cdot n'h' = \underbrace{nhn'h^{-1}}_{\in N} \cdot \underbrace{hh'}_{\in H}$, hence everything is covered by

$$\begin{aligned} \varphi : H &\to \mathrm{Aut}(N) \\ h &\mapsto \varphi_h \end{aligned}$$

where $\varphi_h(n) = hnh^{-1}$ in G. Also $(nh)^{-1} = h^{-1}n^{-1} = (h^{-1}n^{-1}h)h^{-1} = \varphi_{h^{-1}}(n^{-1})h$.

Construction Build "G" on the set $N \times H$.

Associativity:

$$\begin{aligned} ((n,h)(n',h'))(n'',h'') &= (n\varphi_h(n'), hh')(n'',h'') \\ &= (n\varphi_h(n')\varphi_{hh'}(n''), hh'h''), \\ (n,h)((n',h')(n'',h'')) &= (n,h)(n'\varphi_{h'}(n''), h'h'') \\ &= (n\varphi_h(n'\varphi_{h'}(n'')), hh'h''). \end{aligned}$$

To show: $\varphi_h(n')\varphi_{hh'}(n'') = [\varphi_h(n'\varphi_{h'}(n'')) = \varphi_h(n')\varphi_h(\varphi_{h'}(n''))]$,
$\Longleftarrow \varphi_{hh'}(n'') = \varphi_h(\varphi_{h'}(n''))$. φ is a homomorphism $\Longrightarrow \varphi_{hh'} = \varphi_h \circ \varphi_{h'}$.✓

Unit:

$$\begin{aligned} (1_N, 1_H)(n,h) = (\varphi_1(n), h) = (n,h), \because \varphi_1 = 1_{\mathrm{Aut}(N)} = \mathrm{id}, \\ (n,h)(1_N,1_H) = (n\varphi_h(1_N), h) = (n,h), \because \varphi_h(1_N) = 1_N. \end{aligned}$$

Inverse: omitted.

Remains to show both embeds are homomorphisms (omitted), and

$$(n, 1_H)(1_N, h) = (n\varphi_1(1_N), h) = (n,h).$$

To show $N \lhd N \rtimes_\varphi H$, it suffices to show that

$$\begin{aligned} (1_N, h)(n, 1_N)(1_N, h)^{-1} &= (1_N, h)(n, 1_H)(1_N, h^{-1}) \\ &= (1_N, h)(n, h^{-1}) = (\varphi_h(n), 1_H) \in N. \end{aligned}$$

□

Special case: Take $\varphi_h = \mathrm{id}_N, \forall h \in H \Longrightarrow$ get $N \times H$.

Internal version Given a group G and subgroups $N, H \leqslant G$. When can we identify G with $N \rtimes_\varphi H$ for suitable φ s.t. $N \rtimes_\varphi H \simeq G$?

Proposition Assume $N \lhd G, G = NH, N \cap H = \{1\}$. Consider $\mathrm{Ad} : H \to \mathrm{Aut}(N)$, $\mathrm{Ad}_h(n) = hnh^{-1}$. Then $\Phi : N \rtimes_{\mathrm{Ad}} H \simeq G, (n,h) \mapsto nh$ with notation $G = N \rtimes H$.

Proof Φ is a homomorphism:

$$\begin{aligned} \Phi((n,h)(n',h')) &= \Phi(n\mathrm{Ad}_h(n'), hh') \\ &= n\mathrm{Ad}_h(n')hh' = nhn'h' = \Phi((n,h))\Phi((n',h')). \end{aligned}$$

Φ is surjective as $G = NH$.

$\Phi((n,h)) = 1 \Longleftrightarrow nh = 1 \Longleftrightarrow n = h^{-1} \in N \cap H = \{1\}$. Hence Φ is bijective. □

Examples

- Let $n > 1, \tau := (i\ j) \in S_n$. Then:
  - $S_n = A_n \rtimes \langle\tau\rangle$,
    $\because A_n \cap \langle\tau\rangle = \{\mathrm{id}\}, A_n = \ker(\mathrm{sgn}) \lhd S_n$.
    $\forall \sigma \in S_n$, if $\sigma \notin A_n$ then $\mathrm{sgn}(\sigma) = -1 \Longrightarrow \sigma\tau \in A_n, \sigma \in A_n\tau$. Thus $S_n = A_n\langle\tau\rangle$.
  - Note that $n > 2 \Longrightarrow S_n \not\simeq A_n \times \mathbb{Z}/2\mathbb{Z}, \because Z(S_n) = \{\mathrm{id}\}, Z(A_n \times \mathbb{Z}/2\mathbb{Z}) \supset \mathbb{Z}/2\mathbb{Z}$.
- Dihedral groups / 二面体群

  $$\mathrm{SO}(2) \lhd \mathrm{O}(2) := \{\text{orthogonal trans. in } \mathbb{R}^2\}$$

  Let $n \geq 3, \mathrm{D}_{2n} := \{T \in \mathrm{O}(2) \mid T(\text{regular } n\text{-gon}) = \text{regular } n\text{-gon}\}$
  $\Longrightarrow \mathrm{D}_{2n}$ is a subgroup of $\mathrm{O}(2)$.

  Identify $\mathbb{R}^2$ with $\mathbb{C}$, and the regular n-gon has vertices $1, \zeta, ..., \zeta^{n-1}$ with $\zeta = e^{\frac{2\pi i}{n}}$.

  - $\sigma :=$ rotation $R\left(\frac{2\pi}{n}\right)$ on $\mathbb{R}^2 =$ multiplication by ζ on $\mathbb{C} \Longrightarrow \sigma \in \mathrm{D}_{2n}, \mathbb{Z}/n\mathbb{Z} \simeq \langle\sigma\rangle \leqslant \mathrm{D}_{2n}$.
  - $\tau :=$ reflection w.r.t. the X-axis on $\mathbb{R}^2 =$ conjugation on $\mathbb{C} \Longrightarrow \tau \in \mathrm{D}_{2n}, \mathbb{Z}/2\mathbb{Z} \simeq \langle\tau\rangle \leqslant \mathrm{D}_{2n}$.
  - $\tau\sigma^a\tau^{-1} = \tau\sigma^a\tau = \sigma^{-a}$. Can check over $\mathbb{C}$.

  Claim: $\mathrm{D}_{2n} = \langle\sigma\rangle \rtimes \langle\tau\rangle \simeq \mathbb{Z}/n\mathbb{Z} \rtimes_\varphi \mathbb{Z}/2\mathbb{Z}$ where $\varphi(1 + 2\mathbb{Z})$ maps $a + n\mathbb{Z}$ to $-a + n\mathbb{Z}$.

  Proof

  - $\mathrm{D}_{2n} \cap \mathrm{SO}(2) = \langle\sigma\rangle, \because$ any rotation $\in \mathrm{D}_{2n}$ must send 1 to $\zeta^a, a \in \mathbb{Z}/n\mathbb{Z} \Longrightarrow$ it is σ^a.
  - $\mathrm{SO}(2) \lhd \mathrm{O}(2) \Longrightarrow \langle\sigma\rangle \lhd \mathrm{D}_{2n}$.
  - $\tau \notin \mathrm{SO}(2)$, hence $\langle\sigma\rangle \cap \langle\tau\rangle = \{\mathrm{id}\}$.
  - $\tau\sigma^a\tau^{-1} = \sigma^{-a} \Longrightarrow$ the description of φ.
  - Remains to show $\mathrm{D}_{2n} = \langle\sigma\rangle\langle\tau\rangle$.

    Let $g \in \mathrm{D}_{2n}, g(1) = \zeta^k \Longrightarrow \left(\sigma^{-k}g\right)(1) = 1 \in \mathbb{C}$.

    In $\mathbb{C}$ we have $\langle 1\rangle^\perp = \langle \mathrm{i}\rangle \Longrightarrow \sigma^{-k}g$ has matrix $\begin{pmatrix} 1 & \\ & \pm 1 \end{pmatrix}$.

    - "+" $\Longrightarrow \sigma^{-k}g = \mathrm{id}$ in $\mathrm{D}_{2n}, g = \sigma^k$.
    - "−" $\Longrightarrow \sigma^{-k}g = \tau$ in $\mathrm{D}_{2n}, g = \sigma^k\tau$.

  □

  Remark For any $n \geq 1$, we can define $\mathrm{D}_{2n} = \mathbb{Z}/n\mathbb{Z} \rtimes_\varphi \mathbb{Z}/2\mathbb{Z}$ where $\varphi(1 + 2\mathbb{Z})$ maps $a + n\mathbb{Z}$ to $-a + n\mathbb{Z}$. In this sense $\mathrm{D}_2 = \mathbb{Z}/2\mathbb{Z}, \mathrm{D}_4 = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.